

Course Code: CHY1008	Course Title: Basic Chemistry and Environmental Studies	T P C	3	2	4
Version No.	1.0				
Course Pre-requisites / Co-requisites	None				
Anti-requisites (if any)	CHY1009				
Objectives	1. To impart basic concepts of chemical sciences and environmental studies. 2. To get exposure for practical applications of engineering materials. 3. To provide understandings about environment systems and their functions. 4. To present the environmental pollutions and their abatement aspects.				
CO's Mapping with PO's					
Course Outcomes	Course Outcome Statement	PO's			
CO1	Understand comprehensive concept of chemical energy in molecular interactions and transformations.	PO1, PO2			
CO2	Get familiar with engineering materials, nanomaterials, polymers, and their applications in our life.	PO6, PO11			
CO3	Understand natural systems, ecological concepts, biodiversity, and environmental problems and their challenges.	PO2			
CO4	Realize the aspects of environmental hazards, sustainable developments, ethics, and laws.	PO6, PO7			
		TOTAL HOURS OF INSTRUCTIONS: 45			
Module 1	Basic Chemistry and Environmental Studies: Introduction				2 Hours
Chemistry – the central science; Importance, scope and main domains of chemistry; Chemistry in everyday life; Chemistry – A potential foundation to revolutionize technology, industry, energy, healthcare and society. Universe, earth, ecology and environment; Principles and objectives of environmental education; Earth & ecosystems; Ecological principles; Sustainability of natural resources, biodiversity, environmental quality and human societies.					
Module 2	States of Matter, Thermodynamics, Kinetics and Catalysis				9 Hours
States of Matter: Gas, liquid, solid and plasma states. Solid-gas interface - Freundlich and Langmuir adsorption isotherms. Applications of adsorbents.					
Thermodynamics: Laws of thermodynamics. Thermodynamic principles used in air cooler, air conditioner, refrigerator, heat pump, heat engine and IC engine systems.					
Kinetics: Rate law and rate equation (zero, first and second orders). Effect of temperature on rate. Kinetics of ozone layer depletion.					
Catalysis: Types of catalysis – Homogeneous and heterogeneous catalysis. Applications: Fischer Tropsch synthesis, Haber-Bosch process and Three-way automotive catalysts.					
Module 3	Electrochemistry and Chemical Sensors				8 Hours
Electrochemistry: Electrode potential, Nernst equation, types of electrochemical cells, batteries and fuel cells. Theories of corrosion, Anodic and cathodic protection, protective coatings, corrosion inhibitors. Electroplating, electro-machining and electro-winning.					
Chemical Sensors: Components, functions, pH electrode, glucose biosensor, semi-conductor gas sensors.					

Module 4	Engineering Materials	8 Hours
Bulk solids: Network, ionic, molecular, amorphous and crystalline solids. Glass, ceramics, composites, conducting and semiconducting solids, Band theory - Conductivity of solids, optoelectronic materials, OLED. Nanomaterials: Nanoscopic structures - nanospheres, nanotubes, fullerene, and graphene. Nano structuring - 3D Bone void fillers, carbon black for tires, nanocomposites for tennis balls, hard disk drives, magnetic tapes, Ferro fluids, Nanomachines (valves, propellers, robots), <i>etc.</i> Polymers: Thermoplastics, thermosets, elastomers and fibers. Structure-property relations. Properties and applications of polymers – PVC, ABS, Bakelite, Nylon 66, silicones. Conducting and biodegradable polymers. Shrink-wrap polymer films. Phenolic billiard balls.		
Module 5	Ecosystems, Biodiversity, Natural Resources, and Their Human Impacts	6 Hours
Ecosystems: Types, structure and functions of ecosystems. Biogeochemical cycles and human activity impacts. Biomes, estuaries and ecological niches. Biodiversity, Natural Resources, and Their Human Impacts: Types, hotspots, threats and protection of biodiversity. Water, forest, and land resources. Conservation and equitable use of resources for sustainable lifestyles. Natural and man-made environmental disasters – Desertification, Landslides, Forest fires, Tsunamis, Aral sea crisis, Permafrost melting, Coral reef destructions, Floods, Bhopal gas accident, <i>etc.</i> GM Crops and foods. Natural pest control. Organic farming – Sikkim, a total organic farming state of India.		
Module 6	Pollutants and Waste Disposal Methods	6 Hours
Pollutants in Atmosphere: The greenhouse effect, smog formation, stratospheric ozone destruction, acid rain. Pollutants in Hydrosphere: Oxygen-demanding materials, eutrophication, oil and grease spills. Pollutants in Lithosphere: Transport, mobility, and partitioning of pollutants in soils. Pollutants in Biosphere: Classification of hazardous substances, toxicity factors of chemical substances. Waste Disposal Methods: Gas pollutant treatment methods using cyclone separators, bag filters, ESP. Primary, secondary and tertiary effluent treatment methods. Solid waste disposal methods.		
Module 7	Environmental Hazards, Sustainability, Ethics and Laws	6 Hours
Types of environmental hazards. Risk and risk management. Outburst of pandemics – Discussions based on environmental damages and disruptions. Documentary films/videos: An Inconvenient Truth, Plastic Oceans, Invisible Ocean - Plankton & Plastic, Soil-not-Oil, <i>etc.</i> Urban environmental problems. Consumerism and waste products. Sustaining human societies: Economics, environment, policies and education. Environmental ethics and laws. Permaculture – Design principles, Ethics and Applications. Environmental Impact Assessment (EIA) – Predication and evaluation processes of inter-related socio-economic, cultural, and human-health impacts due to human activities.		
Textbooks: 1. M. S. Silberberg, <i>Chemistry: The Molecular Nature of Matter and Change</i>, 9th Edition, McGraw-Hill Education, 2021. 2. G. T. Miller and S. E. Spoolman, <i>Living in the Environment</i>, 20th Edition, Cengage learning, Inc., 2021.		
Reference Books: 1. J. E. McMurry, R. C. Fay and J. K. Robinson, <i>Chemistry</i>, 8th Edition, Pearson Education, Inc., 2020. 2. J. S. Gaffney and N. A. Marley, <i>General Chemistry for Engineers</i>, Elsevier, Inc., 2018. 3. L. S. Brown and T. A. Holme, <i>Chemistry for Engineering Students</i>, 3rd Edition, Cengage learning, 2015. 4. J. Jenata, <i>Principles of Chemical Sensors</i>, Plenum Press, 1989. 5. Anne E. Marteel-Parrish and Martin A. Abraham, <i>Green Chemistry and Engineering: A Pathway to Sustainability</i>, AIChE and John Wiley & Sons, Inc., 2014. 6. David M. Hassenzahl, Mary Catherine Hager and Linda R. Berg, <i>Visualizing Environmental Science</i>, 5th Edition, John Wiley & Sons, Inc., 2017. 7. W. P. Cunningham and M. A. Cunningham, <i>Principles of Environmental Science: Inquiry & Application</i>, 9th Edition, McGraw-Hill Education, 2020. 8. W. P. Cunningham and M. A. Cunningham, <i>Environmental Science: A Global Concern</i>, 14th Edition,		

McGraw-Hill Education, 2018.													
9. Daniel B. Botkin and Edward A. Keller, <i>Environmental Science: Earth as a Living Planet</i> , 8 th Edition, John Wiley & Sons, Inc., 2014.													
10. Thorsten Gressling, <i>Data Science in Chemistry: Artificial Intelligence, Big Data, Chemometrics and Quantum Computing with Jupyter</i> , Walter de Gruyter GmbH, Berlin, 2021.													
11. Emily Gertz and Patrick Di Justo, <i>Environmental Monitoring with Arduino</i> , O'Reilly Media, Inc., 2012.													
Laboratory Experiments													
1. Introduction to the basic handling techniques of chemistry lab instruments 2. Determination of concentration of an unknown acid solution by pH-metry 3. Determination of concentration of an unknown base solution by conductometry 4. Determination of heat of neutralization using solution calorimeter 5. Estimation of heat of solution of KNO ₃ in water by crystallization at different temperatures 6. Enthalpy change determination of an oxidation reaction using a bomb calorimeter 7. Determination of rate constant of hydrolysis of ethyl acetate using conductivity meter 8. Construction of Daniel cell and monitoring its voltage and current 9. Estimation of concentration of bleaching powder by colorimetry 10. Estimation of water quality parameters by water quality analyser 11. Estimation and removal of hardness of water by complexometry and ion exchange resin 12. Estimation of dye concentration in wastewater by UV-VIS Spectrometer 13. Detection of adulteration in food materials – I (oils, fats, milk, honey, beverages. Ref: fssai.gov.in/dart/) 14. Detection of adulteration in food materials – II (food grains, sugar, spices, fruits, vegetables)													
Seminar/Assignment/Project Topics													
Green Chemistry and Green Engineering: Principles and applications. Chemistry and Data Science: Computational chemistry, DNA digital data storage systems, Data visualization and analysis, etc. Environmental Data Science & Statistics: Computing techniques to analyze complex long-term ecological monitoring studies and environmental issues. Green Computing: Sustainable approach in the design, manufacture, use, and disposal of IT resources in an efficient manner for a minimal impact on the environment. Organizations for Nature and Environment: CSE (Centre for Science and Environment), WWF, Greenpeace, Sierra Club Foundation, Environmental Defense Fund, Nature Conservancy, Earthjustice, Oceana, Sea Legacy etc.													
Mode of Evaluation	Continuous Assessment (CATs, FAT, Assignments, Quizzes etc.) <table border="1"> <tr> <td>CAT-1</td><td>20 %</td></tr> <tr> <td>CAT-2</td><td>20 %</td></tr> <tr> <td>FAT</td><td>20 %</td></tr> <tr> <td>LAB</td><td>25 %</td></tr> <tr> <td>Quiz/Assignment</td><td>15 %</td></tr> <tr> <td></td><td>100</td></tr> </table>	CAT-1	20 %	CAT-2	20 %	FAT	20 %	LAB	25 %	Quiz/Assignment	15 %		100
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Recommended by the Board of Studies on	4 th BoS Chemistry (08-11-2021)												
Date of Approval by the Academic Council	7 th Academic Council (09-11-2021)												
Syllabus Prepared by	Dr. K. Selvakumar, Dr. C. Rajesh, Dr. R. Paramasivam												